

The current course slides are static. It is hard for students to understand. There are many good dynamic demos, but students have to check out the link then come back to slides. This is distracting. I intend to put them all together so students can see all core examples in one site with intuitive animations and workflows of each concept. This would make the learning more interactive and in higher efficiency.

Design choices. Single slide-deck site (vanilla HTML/CSS/JS, no build step) with one concept per viewport. Q/K/V color-coded the same way across all 10 slides (purple/orange/green). Every animation step has a plain-English narration paired with the math. A clickable side step-list lets students jump anywhere instantly.

Why it's effective. Intuition comes before symbols: slides 1–2 motivate attention with a real ambiguity ("the animal didn't cross the street because it was too tired") and a YouTube-search analogy before any Q/K/V appears. Softmax, dot products, and sinusoidal positional encoding are all computed live in JS, so every percentage on screen is the real number.

Iterations. ~12 rewrites, mostly about pacing and clarity: bumped fonts three times; slowed all animations behind a tunable SPEED multiplier; made delays interruptible so arrow keys feel instant; added the step list after wanting to rewind; redrew the architecture twice to include residual connections and LayerNorm; fixed an argmax-self bug where narration said "X pays attention to X."

Anything else. Keyboard-first (↑↓ slides, ←→ steps, P pause, ? help). The  $\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d_k})V$  equation on the final slide is the only formula. Everything builds toward it.